ANTARIKSH DAS | 23CS10086

Indian Institute of Technology Kharagpur

B.Tech.(Hons.) in COMPUTER SCIENCE AND ENGINEERING

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EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	B.Tech in Computer Science and Engineering	Indian Institute of Technology Kharagpur (IIT KGP)	9.60/10
2023	AISSCE (CBSE XII)	Sudhir Memorial Institute, Madhyamgram	97.8%
2021	AISSE (CBSE X)	Indian School Al Wadi Al Kabir, Oman	98.2%

INTERNSHIPS

Systems Intern | Quadeye Securities LLP **May 2026 - July 2026**

- **Discovered and fixed critical market data connector bugs across 3 crypto exchanges**, including sequence-number and timestamp parsing, faulty dropping of L2 updates, SBE trade-price parsing errors, faulty uptick logic (price & size), duplicate trade emission, incorrect snapshot-in-incremental-feed handling.
- Built **C++ connection-merging instrumentation** and an **end-to-end self-orchestrating framework** for **granular feed comparison** through latency analysis, contribution metrics, book sync, desync/recovery profiling, trade packet matching, update distribution, and exchange vs matching-engine consistency.
- Designed an **AI-driven analysis engine** for automated inferencing, anomaly detection and diagnosis, and **actionable connector-tuning suggestions** from raw feed analysis output, including correctness checks, minimum-information validation, and independent subagent verification against a knowledgebase.
- Performed exhaustive **feed analysis** across **9 major crypto exchanges**, to identify redundant/ critical feeds as well as unreliable and slow feeds and to test the impact of connection merging. Optimization work on Gate.io alone delivered an **8.5x speedup in runtime**.
- **Migrated Gate.io market data parsing** to **JSON-struct parsing**, **reducing parsing overhead by 83%** and automatically adapting to any change in schema.
- Engineered a **schema-agnostic, generic CSV/directory charting tool** on **Rust**, supporting **arbitrary column counts/orientations**, **multi-metric multi-axis** plotting, multiple aggregation methods, **statistical transformations**, shareable schemas, **automatic schema detection**, and fully generic filtering.
- Built a **GUI-driven CLI tool** on **Python** for **daily sanity-run management** to fetch and categorize sanity stats, generate **reports**, produce **intelligent rerun plans** based on trends, configurable **rerun orchestration** (with optional code/config patch), automated sanity-data diffing and **dashboard synchronization**.
- Developed an **AI-based historical sanity analysis workflow** (single or multi-exchange) to **detect and analyse failing sanities**, isolate **failure onset causes** like code changes, infra failures, tick-size changes, or codebase bugs, with a **deep RCA** engine featuring **DB-based learning**, and a dense knowledgebase.
- **Built automated runtime profiling infrastructure** to identify connector bottlenecks, market data processing overheads, and performance anomalies.
- Implemented a **generic SBE market data decoder** capable of parsing Simple Binary Encoding (SBE) feeds across **heterogeneous exchange schemas**.

PROJECTS

High-Frequency Trading Exchange and Market Simulation | First Place, General Championship Open Soft | repo **April 2026**

- Built a price-time priority **matching engine** (**p50 latency of 180ns**) with support for market/limit orders, partial fills, cancellations, and order modifications.
- Engineered a **cache-efficient limit order book** using **intrusive Red-Black Trees**, **O(1)** lock-free FIFO queues, **slab allocators**, and **zero-allocation hot paths**.
- Designed a **lock-free event-driven architecture** using **single-writer** execution, **SPMC ring buffers**, **zero-copy non-blocking** event propagation.
- Ensured exchange reliability with Write-Ahead Logging (**WAL**), deterministic replay, audit logs, **Merkle**-based state verification, and crash recovery.
- Designed a **binary protocol** supporting **low-latency** order submission, acknowledgements, execution reports, and market data streaming.
- Created **market-making** and **alpha bots** using **OBI**, micro-price, inventory-aware quoting, EMA, adaptive risk management, GBM based market simulation.
- Built a **real time, low latency trading terminal** with OHLC candlesticks, technical indicators and high frequency order book streaming via web sockets.

Quantitative Research: High-Frequency Alpha Discovery and Validation | repo **April 2026**

- Built an event-driven quant **backtesting engine reconstructing a 10-level LOB**, computing **PCA-combined multi-level Order Flow Imbalance** (OFI).
- Modeled order arrivals as a self-exciting **multivariate Hawkes Process (MLE)** and computed **VPIN** to forecast volatility bursts and informed-flow toxicity.
- Trained **XGBoost** on PCA-OFI, Hawkes intensity and spread features to predict sub-second mid-price returns, using **SHAP** to validate LOB-driven alpha.
- Validated alpha via **Combinatorial Purged Cross-Validation** with embargoing, computing **Deflated Sharpe Ratio** and **Probability of Backtest Overfitting**.
- Modeled **Almgren-Chriss market impact** and queue-position adverse selection to derive **AUM capacity curves** bounding strategy scalability.
- Applied **Johansen cointegration** testing and fit a **VECM** to extract adjustment speeds, deriving dynamic hedge ratios for multi-asset statistical arbitrage.

Embedded Distributed Peer-to-Peer DBMS | Course Project | repo *Under Supervision of Prof. Pabitra Mitra and Prof. Pralay Mitra* **March 2026**

- Engineered a **DBMS core** with fixed-page storage, slotted pages, buffer pool, free-space manager, catalog, record manager, and typed value system.
- Added support for **DDL, DML, filtering, grouping, aggregation, ordering, joins**. Built **B-tree and hash indexes** for fast lookups, achieving **6× faster queries**.
- Created an **SQL query processor** comprising lexer/tokenizer, recursive-descent parser, AST, validator, rule-based optimizer, planner, and execution enine.
- Designed an **ACID-compliant transaction layer** using two-phase locking (**2PL**), write-ahead logging (**WAL**), and ARIES-style crash recovery.
- Implemented **Hybrid Logical Clock (HLC)-based MVCC**, row version history, and deterministic conflict resolution for coordinator-free convergence.
- **Designed and implemented a custom P2P network protocol** supporting snapshot synchronization, version exchange, and automatic conflict resolution.
- Built a **multi-transport networking stack** using **TCP** snapshots, **UDP broadcast** peer discovery, **WebSocket relay**, and auto-sync with **no cloud dependency**.
- Implemented **binary message serialization, connection management, synchronization scheduling, fault-tolerant state exchange, and CSV import/export**.

High-Performance Reliable Transport Protocol | Course Project *Under supervision of Prof. Sandip Chakraborty* **March 2026**

- **Built a reliable transport protocol over UDP** with sliding-window ARQ, adaptive RTT/RTO estimation, selective retransmission, and congestion control.
- **Designed a lock-free, io_uring**-based **event-driven runtime** using **slab allocators** and **zero-copy packet paths** to achieve **deterministic, low-latency I/O**.
- **Built a configurable network simulator and deterministic replay harness** to benchmark protocol correctness and tail-latency under high throughput.

AWARDS AND ACHIEVEMENTS

- **Ranked 14** in Codeforces Round 995 and **17** in Codeforces Round 993. Achieved **rating of 1796** and the **Expert** tag (Handle -TheDarkKnight777).
- Earned a **department change** by scoring a **GPA of 9.89** (end of 1st year) and ranked in the **top 1%** among all students who applied.
- Authored a **research paper** published in **IEEE Symposium on Security and Privacy 2025** which received **2nd place** at **CNIL Award for Privacy Protection**.
- Secured **AIR 1817** in JEE Main 2023 out of 11,13,325 students with **100 percentile in physics**, and **AIR 3465** in JEE Advanced 2023.
- Achieved **General Merit Rank(GMR) 28** in the West Bengal Joint Entrance Examination (WBJEE) 2023
- Awarded the prestigious **Kishore Vaigyanik Protsahan Yojana (KVPY)** (2021-22) **SA** Scholarship with **All India Rank 812**.
- Ranked among the **Top 4** teams in the **Kharagpur Data Science Hackathon 2025**, competing against **1,500+** teams **nationwide**.
- Was a **member of the Gold Winning contingent** at **Inter IIT Tech Meet 13.0**.
- **Ranked second** out of the 50 finalist teams at **Overnite 2025**, a **competitive programming** event conducted by Kshitij, IIT Kharagpur.

SKILLS AND EXPERTISE

- **Languages:** *Proficient :* C | C++ | Python | Rust | JavaScript | Typescript | Bash scripts, *Familiar :* Java
- **Frameworks, Tools and Databases :** Git | Docker | Flask | Node.js | Express | Pandas | TensorFlow | Matplotlib | Scikit-Learn | React | Vercel | Next.js | DigitalOcean | Socket.IO | Tailwind CSS | MongoDB | MySQL | PostgreSQL
- **Other Skills:** Machine Learning | Data Analysis | API Development | Quantitative Trading | Full Stack Development | Competitive Programming

COURSEWORK INFORMATION

- **Computer Science :** Algorithms - II | Systems Programming | Software Engineering | Computer Networks | Operating Systems | Database Management Systems | Compilers | Computer Organisation and Architecture | Machine Learning | Deep Learning
- **Mathematics :** Probability and Statistics | Advanced Calculus | Linear Algebra | Numerical and Complex Analysis | Discrete Structures